

equilibrium, necessitating fast kinetics, thereby limiting applicability to high temperatures. Developing simpler material selection criteria for predicting the space charge properties of oxide-based heterointerfaces would go a long way toward providing a practical framework for engineering such composite systems.

A previous study on surface-infiltrated heterostructures by the authors demonstrated that the acidity of binary oxides acts as a sensitive descriptor for predicting the surface oxygen exchange rate (k_{chem}) in MIEC electrode²¹. This was attributed to near-surface electron depletion or accumulation, driven by the modulated space charge potential at the heterointerface, which correlated the relative Smith acidity of the surface infiltrated binary oxides relative to that of the host MIEC electrode^{21–24}. Building on this insight, the Smith acidity scale, or equivalently the relative work functions of the oxides, serves as a powerful descriptor of electronic and ionic properties induced by local space charges formed at a broader category of heterointerfaces. This serves to provide a rational and useful guideline for materials selection that elucidates the relation between local space charge and conductivity. We furthermore note that the Smith acidity is derived in terms of an oxide's tendency to accept an oxygen anion from a basic oxide during a chemical reaction and calculated using enthalpy changes during such reactions. This framework is consistent with the framework derived by Steinbach et al., as mentioned above.

Although changes in the electrical conductivity of porous $\text{Pr}_{0.1}\text{CeO}_{0.9}\text{O}_{2-\delta}$ layers upon acidic/basic binary oxide infiltration were also assessed, they were modest (10–25%)²¹, leading several studies to question the space charge model, despite similar trends being observed in electrode polarization with controlled surface acidity^{17–19}. In part, this was based on their inability to detect corresponding changes in the electrical conductivity of their films and little detectable change in the oxidation state of the mixed valent elements in the MIEC surface. We suspect that the smaller changes in conductivity compared to that of k_{chem} were primarily due to the bulk-dominated character of conventional MIEC electrodes^{25–27}. Given the nanometer-scale dimensions of space charge regions and the low surface coverage (1.5–3%) achieved during the infiltration process, the influence of acidity-controlled heterointerfaces on overall conductivity was limited^{21,28}. These issues therefore necessitate the development of an alternative platform with a much higher surface-to-volume ratio to maximize the influence of surface conduction and examine precisely the impact of heterointerface acidity-mediated space charge on the near-surface electronic properties. Such an approach should serve to unveil the

correlation between modified local space charge potentials induced by surface acidity and the corresponding electrical conductivity changes.

In this study, we successfully demonstrate the marked effects of heterointerface acidity on the electronic properties of an optimized nanostructured platform composed of well-aligned $\text{Pr}_{0.2}\text{Ce}_{0.8}\text{O}_{2-\delta}$ nanowire arrays (PCO_{NA}). The conductivity of PCO_{NA} is demonstrated to vary by three orders of magnitude over the temperature range of 450-650 °C, following changes in relative Smith acidity of infiltrated binary oxides, ranging from basic (Li_2O) to acidic (SiO_2). Moreover, serial infiltration with Li_2O following reductions in electrical conductivity induced initially by SiO_2 , not only recovers, but exceeds the conductivity initially measured by nearly the same factor. We also report, for the first time, and consistent with the above observations, that the relative acidity of insulating substrates (Al_2O_3 vs. MgO) supporting PCO_{NA} can also significantly impact its electronic properties. Finally, we note that in polycrystalline materials, space charge potentials at grain boundaries (GBs) must also be accounted for in rationalizing conductivity changes, along with those associated with the heterointerfaces. This is because cation diffusion from substrates or surface infiltrants into polycrystalline thin films can alter their GB space charge potentials, depending on whether in-diffused species act as donors or acceptors at the specific GB lattice sites they occupy. As a result, the overall conduction pathway governing the electronic properties of PCO_{NA} is determined by the acidity-mediated space charge potentials at both the heterointerfaces and GBs, representing clear and strong evidence for the critical role of local space charges in MIEC electrodes.

Results

Fabrication and characterization of PCO_{NA} as a model platform

Fig. 1a illustrates a cross-sectional view of a PCO_{NA} model platform supported on an insulating substrate to elucidate the role of local space charges induced by modulated surface or substrate acidity on electronic properties. In addition to the bulk conduction (R_1), four other possible conduction pathways within PCO_{NA} have been depicted as conduction through substrate heterointerfaces (R_2), along GBs (R_3), perpendicular to GBs (R_4), and surface heterointerfaces with infiltrants (R_5). Because space charge potentials at heterointerfaces and GBs are affected by the relative acidity of binary oxides and the in-diffusion of cations along GBs sourced from the binary oxides, all these conduction pathways need to be considered. Notably, all other conduction pathways are always in series with conduction perpendicular to the GBs (R_4), so the overall equivalent circuit can be described as shown in **Fig. 1b**.

To achieve a high surface-to-volume ratio to enhance the relative ratio of surface to bulk conduction, PCO_{NA} was fabricated using a solvent-assisted nanotransfer printing method (S-nTP) (**Fig. 1c**), previously reported^{29–31}. This morphology offers readily accessible open structures for the conformal formation of interfaces with both substrates and infiltrants. As shown in the scanning electron microscopy (SEM) images (**Fig. 1d**), PCO_{NA} with widths of 50 nm, pitch of 150 nm, and thickness of 50 nm were fabricated on insulating substrates. Due to the shadowing effect caused by oblique angle deposition³², Pr_{0.2}Ce_{0.8}O_{2-δ} (PCO20) was deposited only on one side of the PMMA, resulting in an unconventional droplet-shaped structure with a tail-like cross-section (see inset). This nanoscale morphology contributes to an increased surface-to-volume ratio, intensifying the influence of surfaces properties compared to that of the bulk, and provides a large number of parallel conductive nanowire channels (estimated to be 50,000) for precise analysis of the in-plane electrical conductivity. The concentration of Pr (20 at%) in the fabricated PCO_{NA} was identical to that of the target source as confirmed by inductively-coupled plasma mass spectrometry (ICP-MS) (**Supplementary Table 1**). The accompanying transmission electron microscopy (TEM) image shows that the PCO20 nanowires are composed of nanoscale grains with diameters ranging from a few up to 20 nm (**Fig. 1e**), and that these nanosized grains result in a rough surface, which amplifies the impact of the surface on properties. While PCO20 is shown to be polycrystalline, most of the crystal planes are observed to consist of CeO₂ (111) planes with d-spacings of 0.31 nm. X-ray diffraction (XRD) spectra also assign the main peak to the (111) plane, although other oriented planes also exist (**Fig. 1f**). Furthermore, the identical positions of peaks appearing in fabricated